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Web Design

Lecture 11 - JavaScript & HTML DOM



7.1. Document Objects Model (DOM)

7.2. Event handling

7.3. Form validation

7.4. AJAX



HTML DOM

DOM methods

DOM documents

DOM elements

DOM changing HTML

DOM changing CSS



The DOM is a W3C (World Wide Web Consortium) standard.

"The W3C Document Object Model (DOM) is a platform and language-neutral interface that allows programs and scripts to dynamically access and update the content, structure, and style of a document."



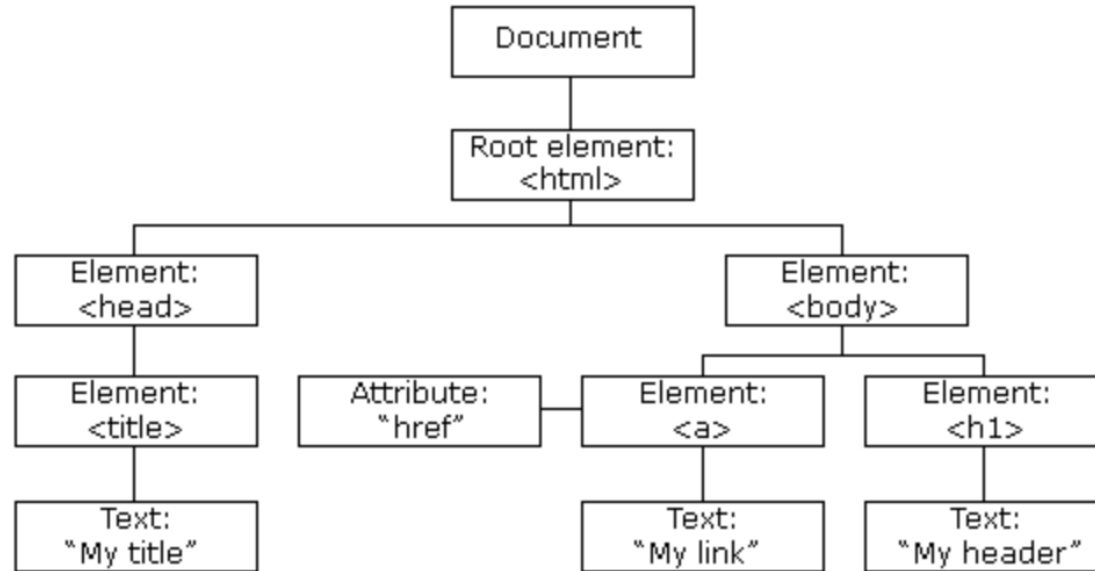
The HTML DOM is a standard object model and programming interface for HTML. It defines:

- The HTML elements as objects
- The properties of all HTML elements
- The methods to access all HTML elements
- The events for all HTML elements

In other words: The HTML DOM is a standard for how to get, change, add, or delete HTML elements.



With the HTML DOM, JavaScript can access and change all the elements of an HTML document.





With the object model, JavaScript gets all the power it needs to create dynamic HTML:

- JavaScript can change all the HTML elements in the page
- JavaScript can change all the HTML attributes in the page
- JavaScript can change all the CSS styles in the page
- JavaScript can remove existing HTML elements and attributes
- JavaScript can add new HTML elements and attributes
- JavaScript can react to all existing HTML events in the page
- JavaScript can create new HTML events in the page



HTML DOM methods are actions you can perform (on HTML Elements).

HTML DOM properties are values (of HTML Elements) that you can set or change.

A property is a value that you can get or set (like changing the content of an HTML element).

A method is an action you can do (like add or deleting an HTML element).



In the example above, **getElementById** is a *method*, while **innerHTML** is a *property*.

```
<html>
<body>

<p id="demo"></p>

<script>
document.getElementById("demo").innerHTML = "Hello World!";
</script>

</body>
</html>
```



Finding HTML Elements

Method	Description
<code>document.getElementById(<i>id</i>)</code>	Find an element by element id
<code>document.getElementsByTagName(<i>name</i>)</code>	Find elements by tag name
<code>document.getElementsByClassName(<i>name</i>)</code>	Find elements by class name



Changing HTML Elements

Property	Description
<i>element.innerHTML = new html content</i>	Change the inner HTML of an element
<i>element.attribute = new value</i>	Change the attribute value of an HTML element
<i>element.style.property = new style</i>	Change the style of an HTML element
Method	Description
<i>element.setAttribute(attribute, value)</i>	Change the attribute value of an HTML element



Adding and Deleting Elements

Method	Description
<code>document.createElement(<i>element</i>)</code>	Create an HTML element
<code>document.removeChild(<i>element</i>)</code>	Remove an HTML element
<code>document.appendChild(<i>element</i>)</code>	Add an HTML element
<code>document.replaceChild(<i>new</i>, <i>old</i>)</code>	Replace an HTML element
<code>document.write(<i>text</i>)</code>	Write into the HTML output stream



Adding Events Handlers

Method	Description
<code>document.getElementById(<i>id</i>).onclick = function() {<i>code</i>}</code>	Adding event handler code to an onclick event



Practice:

https://www.w3schools.com/js/js_html_dom_elements.asp

Exercise:

https://github.com/sinhvtr/web_design_is19a/blob/main/JavaScript%20Exercises/Lecture%2011%20-%20JS%20%26%20DOM/Lecture11.html

1. Tìm tất cả phần tử div chứa đoạn text thể hiện giá sản phẩm, có bao nhiêu phần tử như vậy
2. In ra nội dung bên trong phần tử div cuối cùng tìm được
3. Tìm tất cả phần tử ul trong trang web này, in ra số lượng phần tử tìm được



Changing HTML Content

The easiest way to modify the content of an HTML element is by using the innerHTML property.

```
document.getElementById(id).innerHTML = new HTML
```

Changing the Value of an Attribute

```
document.getElementById(id).attribute = new value
```

https://www.w3schools.com/js/tryit.asp?filename=tryjs_dom_image

https://www.w3schools.com/js/tryit.asp?filename=tryjs_dom_date



Exercises:

4. Tìm đến phần tử chứa dòng chữ Giỏ hàng, đổi dòng chữ này thành Cart
5. Tìm đến phần tử chứa hình ảnh sản phẩm đầu tiên trên trang chủ, đổi hình ảnh thành 1 ảnh khác bất kỳ



Changing HTML Style

document.getElementById(id).style.property = new style

https://www.w3schools.com/js/tryit.asp?filename=tryjs_change_style

Using Events

Events are generated by the browser when "things happen" to HTML elements:

- An element is clicked on
- The page has loaded
- Input fields are changed

https://www.w3schools.com/js/tryit.asp?filename=tryjs_dom_color2



6. Chuyển tất cả giá sản phẩm thành chữ màu xanh blue, in đậm

7. Tạo sự kiện khi di chuột vào ảnh logo thì width và height của ảnh set lên thành 100

8. Tạo sự kiện khi di chuột ra khỏi ảnh logo thì width và height của ảnh set về 50 như cũ

https://www.w3schools.com/js/tryit.asp?filename=tryjs_dom_animate_3



Reacting to Events

HTML Event Attributes

Assign Events Using the HTML DOM



When a user clicks the mouse
When a web page has loaded
When an image has been loaded
When the mouse moves over an element
When an input field is changed
When an HTML form is submitted
When a user strokes a key

https://www.w3schools.com/js/tryit.asp?filename=tryjs_event_onclick3



To assign events to HTML elements you can use event attributes.

```
<button onclick="displayDate()">Try it</button>
```

https://www.w3schools.com/js/tryit.asp?filename=tryjs_events1

```
<script>  
document.getElementById("myBtn").onclick = displayDate;  
</script>
```



The `addEventListener()` method, e.g.:

```
document.getElementById("myBtn").addEventListener("click", displayDate);
```

The `addEventListener()` method attaches an event handler to an element without overwriting existing event handlers.

Syntax:

```
element.addEventListener(event, function, useCapture);
```

You can easily remove an event listener by using the `removeEventListener()` method.

https://www.w3schools.com/js/tryit.asp?filename=tryjs_addeventlistener_add2



Exercises 11-12 / exercise_2.html

Sample: ***Exercises 11-12 / find_subjects_in_a_list.html***



DOM Navigation

https://www.w3schools.com/js/js_htmlDOM_navigation.asp

DOM Nodes

https://www.w3schools.com/js/js_htmlDOM_nodes.asp



Basic Form Validation

Data Format Validation



HTML form validation can be performed automatically by the browser:

If a form field (fname) is empty, the required attribute prevents this form from being submitted:

```
<form action="/action_page.php" method="post">  
  <input type="text" name="fname" required>  
  <input type="submit" value="Submit">  
</form>
```

https://www.w3schools.com/js/js_validation.asp



HTML form validation can be done by JavaScript.

https://www.w3schools.com/js/tryit.asp?filename=tryjs_validation_js

```
<form name="myForm" action="/action_page.php" onsubmit="return validateForm()" method="post">  
Name: <input type="text" name="fname">  
<input type="submit" value="Submit">  
</form>
```

```
function validateForm() {  
    let x = document.forms["myForm"]["fname"].value;  
    if (x == "") {  
        alert("Name must be filled out");  
        return false;  
    }  
}
```



Exercises 11-12 / exercise_1.html



AJAX is not a programming language.

AJAX is a technique for accessing web servers from a web page.

AJAX stands for Asynchronous JavaScript And XML.

https://www.w3schools.com/js/js_ajax_intro.asp